CHAPTER 1
INTRODUCTION

1.1. Introduction

The turbofan engine, key to modern aircraft propulsion, integrates turbojet principles with a ducted fan to boost efficiency. It uses a gas turbine to generate mechanical energy and a fan to propel air rearwards. Turbofans are classified into high-bypass and low-bypass types: high-bypass engines are favored in commercial aviation for fuel efficiency, while low-bypass engines are used in military fighters for speed and agility.

The combustor, located between the compressor and turbine, is crucial for the combustion process. Among combustor types tubular, tubo-annular, and annular. The annular combustor is most common in modern aero-engines. It features a concentric design with stable combustion, compact size, low pressure drop, and uniform exit temperatures.

Computational fluid dynamics (CFD) analysis of combustors typically uses combustion models such as the k-epsilon, k-omega, and k-omega shear stress transport (SST) models to simulate turbulent combustion.

1.2. Historical Review

For the last two centuries, investigators made many attempts in order to find the best mobile capable of transforming the fuel energy into mechanical work, in a most simple way. Since then, the evolution of the Gas Turbine Engines (GTE) is directly related to the evolution process of the combustor, due to the importance of this in a GTE.

In the 18th century, Sir Isaac Newton theorized that an explosion directed backward could propel an aircraft forward rapidly, based on his third law of motion: for every action, there is an equal and opposite reaction. Since then, various attempts have been made to build an engine based on this principle. The first attempt was by Henry Giffard in 1852, who developed a three-horsepower steam engine to propel his airship. Although it was initially successful, it lacked sufficient power for proper navigation. In

1894, Hiram Maxim's triple biplane, also powered by a steam engine, managed only a brief flight due to the heavy coal-powered engine. These early attempts failed primarily because steam engines were too heavy for flight. The invention of the internal combustion reciprocating engine led to the first powered flight by the Wright Brothers in 1903 and remained the main propulsion method until the late 1930s.

By the time World War II (WWII) approached, military aircraft relied on internal combustion piston engines. However, these engines faced limitations in acceleration and maneuverability due to their size and weight, and propeller efficiency decreased as blade tips neared the speed of sound. These challenges led to the development of a new type of engine, the gas turbine engine, or jet engine.

British pilot Frank Whittle designed and patented the first turbojet engine in 1930, but the first successful jet-powered flight occurred in Germany in 1939, developed by physicist Hans von Ohain. Whittle initially struggled with combustor design issues, such as thermal cracking and coking in the vaporizing tubes and difficulty in controlling fuel flow rates. To overcome these challenges, Whittle replaced the vaporizer tubes with a pressure-swirl atomizer and added a large air swirler, as shown in Figure 1.1, to create a reversal toroidal flow for better mixing of fuel vapor, air, and combustion products.

This revised combustor design, depicted in Figure 1.1, was incorporated into the Power Jets W1 engine, featuring 10 separate tubular combustors in a reverse-flow arrangement to create a compact engine. This innovative configuration enabled the engine to achieve higher heat-release rates and made it possible for the first British turbojet flight on May 15, 1941.

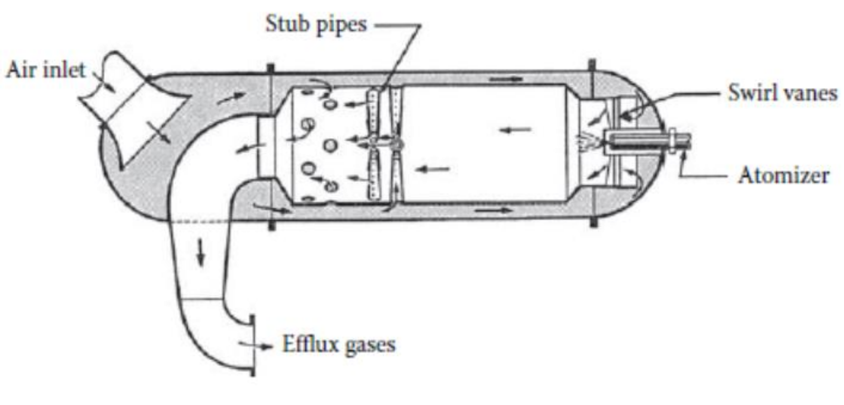

Figure 1.1. Reverse-Flow Atomizer Combustor

Later in Britain, the straight-through combustor was developed and used in the De Havilland Goblin engine, while the first annular combustor was introduced in the Metrovick engine. The annular combustor design featured upstream fuel injectors and downstream dilution air. The upstream fuel injectors allowed longer fuel residence time in the combustion zone for better evaporation, while the downstream dilution air managed the air distribution for combustion and temperature control. However, this design proved unsuitable for aircraft due to the increased weight and vulnerability of the scoops to burnout from high-velocity combustion gases. Additionally, upstream fuel injectors were phased out due to persistent carbon buildup on the atomizer face.

The Jumo 004 and BMW 003 were the only axial-flow turbojet engines produced during WWII. The Jumo 004, developed by Anselm Franz, utilized six-can combustors. Although Franz recognized the benefits of an annular combustor, he chose a can configuration for simpler testing and development. Three of the cans were equipped with spark plugs, and interconnectors ensured ignition across all combustors. Each combustor, designed to burn diesel fuel, employed a pressure swirl atomizer at fuel pressures up to 5.2 MPa. To achieve efficient combustion and a compact flame, primary combustion air was introduced through helical slots, creating a swirl that mixed fuel and air effectively. A cutaway view of the combustor can design is shown in Figure 1.2.

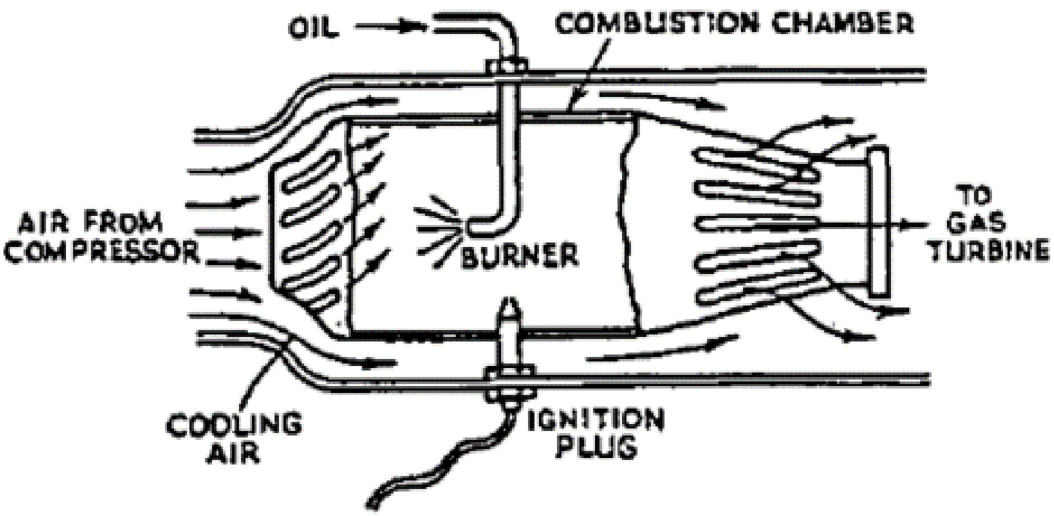

Figure 1.2. Jumo 004 Tubular Combustor

The BMW 003 was the second successful axial-flow turbojet engine used during WWII, despite its development beginning earlier than the Jumo 004. This engine featured an annular combustor with 16 equally spaced downstream pressure atomizers, each surrounded by baffles. Primary combustion air flowed through and around the nozzles to ensure efficient combustion. Dilution air was supplied by 40 scoops on the outer liner, helping to complete combustion and lower the gas temperatures exiting the combustor.

The design provided a low-pressure loss, which improved the engine's efficiency, but it also resulted in a high length-to-diameter ratio, making the engine relatively long. However, the early version of the combustor had a limited operational lifespan of just 25 hours due to the use of mild steel with an aluminum coating that could not withstand the high temperatures inside the combustor. This limitation delayed the engine's potential impact in combat situations.

During WWII, the development of jet engines was hindered by the secrecy surrounding technological advancements. Despite being far ahead in jet engine research, Germany was the only country to deploy a jet-powered combat aircraft, the Messerschmitt Me-262, but its late introduction did not alter the course of the war. The end of the war in 1945 led to the discovery of many wartime technological breakthroughs, which helped to resolve jet engine problems for both Britain and Germany.

Pre-WWII jet engines were inefficient, unreliable, and produced high levels of noise and pollution. However, after the war, these newly discovered technologies greatly enhanced engine performance. Over the next 20 years, jet engines became the standard propulsion system for civil aviation. During this time, companies such as General Electric (GE), Rolls-Royce (RR), and Pratt & Whitney (P&W) developed more efficient jet engines for commercial and military aircraft.

General Electric first used a reverse-flow combustor in its J31 engine, based on Whittle's design, but later transitioned to a more efficient straight-through combustor in the J33, J35, and J47 engines, significantly improving performance. Meanwhile, Pratt & Whitney's J57 engine featured eight tubular combustion liners within an annular casing, with each liner containing a perforated central tube. Fuel was injected through six equally spaced pressure-swirl atomizers, enhancing combustion efficiency and overall engine performance, resulting in greater reliability under various operating conditions.

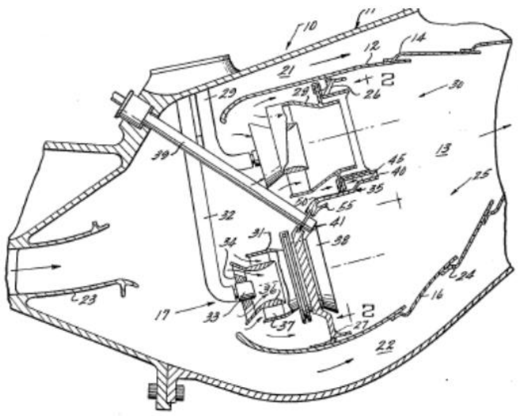

Figure 1.3. An Axial Cross-Sectional View of a Double Annular Combustor

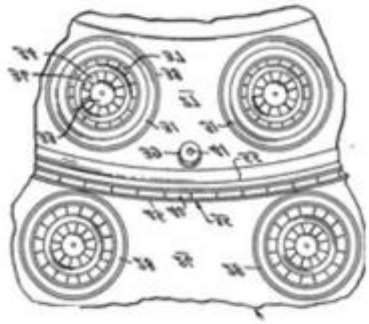

Figure 1.4. A Front View of the Injectors

In the mid-1970s, the need to reduce aircraft emissions spurred significant advancements in engine design. One notable innovation was the development of staged combustion techniques, including the Double Annular Combustor (DAC), which is illustrated in Figures 1.3 and 1.4. The DAC is designed with two concentric stages: an outer pilot stage that operates at low power and low emissions, and an inner main stage that delivers high power while minimizing NOx emissions. Although initially implemented in the 1950s J34 engine, the DAC gained significant attention when General Electric integrated it into the CFM56-5B engine in the 1990s.

In 1995, GE and NASA introduced the Twin Annular Premixing Swirler (TAPS) combustor, as shown in Figure 1.5. The TAPS combustor features a central pilot injector and a concentric outer main injector, providing a flexible fuel management system. At low power settings, the pilot injector manages 100% of the fuel, while at high power, the main injector facilitates lean burning. This design effectively reduces emissions during critical Landing and Take Off (LTO) cycles, addressing environmental concerns while maintaining performance.

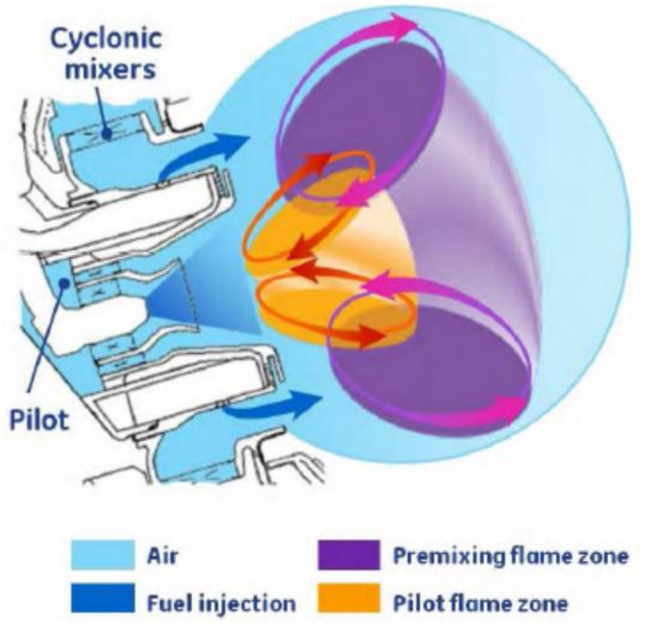

Figure 1.5. TAPS Fuel Injection Concept

Despite modifications over time, the fundamental design of aero-engine combustors has remained largely consistent, evolving gradually rather than through dramatic changes. Consequently, most current aero-engine combustors are similar in size, shape, and appearance, with key components including the diffuser, combustion chamber, fuel injectors, casings, and, in can combustors, interconnectors.

Inside the combustor liner, air distribution is designed to maintain a higher fuel/air ratio in the primary combustion zone compared to other zones. Additional air is introduced downstream for two purposes: to complete combustion in the primary zone and to cool the walls and dilute combustion products to a temperature suitable for the turbine. This careful design ensures efficient combustion while protecting sensitive engine components from excessive heat. Furthermore, the effective management of air distribution plays a critical role in optimizing performance and emissions control.

1.3. Aim and Objectives

This thesis aims to study the annular combustor in high bypass turbofan engines using computational fluid dynamics. It focuses on calculating key combustion parameters, including pressure loss, combustion intensity, efficiency, and fundamentals such as enthalpy of formation and combustion, as well as combustion stoichiometry. The objective is to enhance understanding of annular combustors in high bypass turbofan engines.

1.4. Implementation Program

This thesis has been caried out step by step. Firstly, study on the of aircraft engines and the high bypass turbofan engine. Next, study on annular combustor and combustion behaviors. Finally, case studies on numerical modelling of annular combustor.

1.5. Scope of the Thesis

This thesis is to provide and an essential help for studying of the annular combustor in high bypass turbofan engine and to learn behaviour of the combustion in annular combustor. The main performance of this thesis is to calculate combustion parameters and combustion fundamentals in annular combustor in high bypass turbofan engine.

1.6. Outlines of Thesis

This thesis is organized into six chapters. Chapter one provides the introduction. Chapter two reviews the literature on aircraft engines and combustors. Chapter three covers combustion parameters and fundamentals. Chapter four details the calculation of combustion parameters. Chapter five focuses on the numerical modeling of the annular combustor. The final chapter presents the discussion and conclusion.